

The Complexity of Stationary Diffusion-Driven Instability

Exact Semialgebraic Characterization, NP-Hardness, and Fixed-Species Algorithms

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Abstract

We consider a finite classical mass-action reaction network specified by its indexed source-complex matrix Y and stoichiometric matrix Γ . We ask whether some strictly positive rates and homogeneous equilibrium are stable relative to the stoichiometric compatibility class, while some strictly positive all-species diagonal diffusion matrix creates a nonzero spatial mode with a positive real eigenvalue. We prove an exact characterization: there must exist a positive steady flux v , $\Gamma v = 0$, and a positive reciprocal concentration vector h such that $J = \Gamma \operatorname{diag}(v) Y^T \operatorname{diag}(h)$ is Hurwitz on $\operatorname{im} \Gamma$ and has a negative signed principal minor. This yields a polynomial-size formula in the existential theory of the reals and finite exact certificates on both sides. The decision problem is NP-hard under a polynomial-time reduction from PARTITION. For every fixed species count, however, it is decidable in polynomial bit complexity by projecting the positive flux cone into the fixed-dimensional space of Jacobian factors. Conservation laws are handled without restricting nonzero diffusive modes to the stoichiometric subspace. The hardness construction uses reaction complexes of unbounded molecularity and proves only weak NP-hardness. The results concern weak stationary linear instability; they do not assert a first or transverse crossing, exclude an earlier wave instability, or prove a nonlinear patterned state.

1 Introduction

Turing’s diffusion-driven instability mechanism asks how diagonal diffusion can destabilize a homogeneous equilibrium that is stable without spatial transport [16]. Matrix criteria are well developed when the Jacobian is fixed [13, 19, 1, 10], and interaction-topology studies have identified structural features and surveyed small networks [8, 6, 14]. A reaction network, however, does not specify one Jacobian. Under classical mass action, positive rates and positive equilibria generate a constrained family whose entries are coupled by steady flux and concentration scaling.

This paper studies the corresponding network-wide decision problem. The input is the complete indexed pair (Γ, Y) , not merely a signed interaction graph. We call the network *weakly all-mobile stationary Turing-admissible* if there are strictly positive rate constants, a strictly positive homogeneous equilibrium, and strictly positive species diffusion coefficients such that the homogeneous Jacobian is Hurwitz on the stoichiometric subspace and a nonzero Neumann mode has a strictly positive real eigenvalue.

Our first contribution is an exact reduction of this question to positive steady flux, relative Routh–Hurwitz inequalities, and one negative signed principal minor. The diffusion coefficients, domain scale, original rates, and equilibrium concentrations are eliminated from the destabilization condition. Boolean selector variables encode the principal-set disjunction with polynomial size, proving membership in the existential theory of the reals.

The second contribution is a complexity boundary. Starting from the open-interval matrix-stability construction of Blondel and Tsitsiklis [3], we introduce a row-splitting lift whose stability is unchanged and whose arbitrary positive right scalings can be eliminated by an inertia and determinant-continuity argument. We then realize the entire lifted row family—not just one matrix—as the positive steady-flux image of a classical mass-action network. This gives weak NP-hardness from PARTITION. The reduction has full stoichiometric rank and unit stoichiometric columns, but the source and target molecularities are not uniformly bounded.

The third contribution is tractability at fixed species count. Extreme rays of the nonnegative steady-flux cone are positive circuits supported on at most $n + 1$ reactions. For fixed n they can be enumerated in polynomial time, projected into $\mathbb{R}^{n \times n}$, and converted in fixed dimension to a relative-facet description. Fixed-variable real-algebraic decision then gives a polynomial-time exact algorithm for every fixed n , including arbitrary one-, two-, and three-species reaction lists.

Scope. The word *stationary* here means that the unstable spatial matrix has a positive real eigenvalue. We do not require this to be the first loss of stability along a diffusion path, do not require simplicity or transversality, and do not exclude an earlier complex crossing. A linear Turing instability also does not by itself guarantee a persistent nonlinear pattern [7]. No such conclusion is made.

Relation to structural reaction-network work. Graph criteria for zero-eigenvalue Turing instability under mass action [9], unstable cores and child selections under parameter-rich kinetics [18, 17], and recent sufficient conditions for networks with monomial steady-state parametrizations [5] are closest in subject. Their kinetic classes, quantifiers, or necessity/sufficiency scopes differ from the arbitrary-network classical-mass-action decision problem considered here. The fixed-matrix principal-minor result is used in an exact form and re-proved; it is not presented as the principal novelty.

2 Mass-action Jacobians and conservation laws

Consider indexed reactions

$$y_r \longrightarrow y'_r, \quad r = 1, \dots, m,$$

with $y_r, y'_r \in \mathbb{Z}_{\geq 0}^n$ and $y_r \neq y'_r$. Let

$$Y = [y_1 \ \cdots \ y_m], \quad \Gamma = [y'_1 - y_1 \ \cdots \ y'_m - y_m], \quad S = \text{im } \Gamma.$$

With positive rates k_r , the vector field is

$$f_k(x) = \Gamma r_k(x), \quad r_{k,r}(x) = k_r x^{y_r}.$$

Proposition 2.1 (Positive-equilibrium parametrization). *At a positive equilibrium x^* define*

$$v_r = k_r (x^*)^{y_r}, \quad h_i = (x_i^*)^{-1}.$$

Then $v > 0$, $\Gamma v = 0$, and

$$J(v, h) = \Gamma \text{diag}(v) Y^T \text{diag}(h). \tag{1}$$

Conversely, every $v > 0$ with $\Gamma v = 0$ and every $h > 0$ are realized by

$$x_i^* = h_i^{-1}, \quad k_r = \frac{v_r}{(x^*)^{y_r}}.$$

Proof. The derivative of $k_r x^{y_r}$ at x^* is $v_r y_r^T \text{diag}(h)$, giving (1). The equilibrium condition is $\Gamma v = 0$. The displayed converse rates are positive, recover the flux v , and therefore give both equilibrium and the same Jacobian. $\square$

This parametrization includes zero complexes, parallel indexed reactions, autocatalytic reactions, conservation laws, and arbitrary molecularity.

Proposition 2.2 (Conservation splitting). *Suppose $J(\mathbb{R}^n) \subseteq S$, $J(S) \subseteq S$, and $J|_S$ is Hurwitz. Then*

$$\mathbb{R}^n = S \oplus \ker J, \quad J \sim \begin{pmatrix} J|_S & 0 \\ 0 & 0 \end{pmatrix}.$$

The conservation-induced zero eigenvalues are semisimple and, with $s = \dim S$,

$$\det(\lambda I - J) = \lambda^{n-s} \det(\lambda I_s - J|_S) > 0 \quad (\lambda > 0). \quad (2)$$

Proof. The restriction is invertible, so $J(S) = S$. Since the full image is contained in S , $\text{im } J = S$. Invertibility on S gives $S \cap \ker J = \{0\}$, and rank-nullity gives the direct sum. Both summands are invariant and J vanishes on its kernel. The determinant factorization and positivity follow. $\square$

Choose a rational basis matrix $B \in \mathbb{Q}^{n \times s}$ for S and a rational left inverse C with $CB = I_s$. Then

$$R(v, h) = CJ(v, h)B \quad (3)$$

represents $J|_S$. If $\tilde{B} = BT$, the new representation is $T^{-1}R(v, h)T$; all Hurwitz tests are basis-independent.

For Neumann boundary conditions, a nonconstant Laplacian eigenfunction ϕ has zero spatial mean. Therefore a perturbation $a\phi(x)$ preserves every integrated conservation law for every amplitude $a \in \mathbb{R}^n$. The nonzero-mode amplitude space is the full species space, and its matrix is $J - q^2 D$. It would generally be wrong to restrict this matrix to S : unequal diagonal diffusion need not preserve S pointwise.

3 Fixed-J diagonal damping and the exact characterization

For $I \subseteq [n]$, write

$$c_I(J) = (-1)^{|I|} \det J[I, I], \quad c_\emptyset(J) = 1.$$

Theorem 3.1 (Positive diagonal damping). *For every real square matrix J , the following are equivalent:*

- (i) *some $D \succ 0$ diagonal makes $J - D$ have a strictly positive real eigenvalue;*
- (ii) *$c_I(J) < 0$ for some principal set I .*

If J is Hurwitz, a witnessing set is nonempty and proper.

Proof. Multilinearity in the columns gives

$$\det(D - J) = \sum_{I \subseteq [n]} c_I(J) \prod_{j \notin I} d_j. \quad (4)$$

If every $c_I(J) \geq 0$, then for $\lambda > 0$ the same expansion with $d_j + \lambda$ is strictly positive because the empty-set term is positive. Thus $\det(\lambda I - (J - D))$ cannot vanish at a positive real λ .

Conversely, fix I with $c_I(J) < 0$ and set $d_j = t$ off I and $d_j = t^{-1}$ on I . The exponent of the I -term exceeds that of the K -term by $|I \Delta K|$, so the negative I -coefficient uniquely dominates for large finite t . Hence $\det(D - J) < 0$. The monic characteristic polynomial of $J - D$ is negative at zero and tends to $+\infty$ on the positive real axis, so it has a positive real root. When J is Hurwitz, $c_\emptyset = 1$ and $c_{[n]} = \det(-J) > 0$. $\square$

Set

$$A(v) = \Gamma \operatorname{diag}(v) Y^T.$$

Column scaling gives, for every principal set,

$$(-1)^{|I|} \det J(v, h)[I, I] = \left(\prod_{i \in I} h_i \right) (-1)^{|I|} \det A(v)[I, I]. \quad (5)$$

Equivalently, Cauchy–Binet expands the second factor as

$$(-1)^{|I|} \sum_{\substack{T \subseteq [m] \\ |T|=|I|}} \det(\Gamma_{I,T}) \det(Y_{I,T}) \prod_{r \in T} v_r.$$

Cancellations among reaction selections are therefore retained exactly.

Theorem 3.2 (Exact network characterization). *A finite classical mass-action network is weakly all-mobile stationary Turing-admissible if and only if there exist*

$$v \in \mathbb{R}_{>0}^m, \quad h \in \mathbb{R}_{>0}^n, \quad \emptyset \neq I \subseteq [n],$$

such that

$$\Gamma v = 0, \quad (6)$$

$$R(v, h) \text{ is Hurwitz}, \quad (7)$$

$$(-1)^{|I|} \det A(v)[I, I] < 0. \quad (8)$$

A witnessing set is necessarily proper.

Proof. Given a positive mass-action equilibrium, Theorem 2.1 produces v, h . Homogeneous stability relative to the compatibility class is exactly (7). If a nonzero mode has a positive real eigenvalue, Theorem 3.1 gives a negative signed principal minor of J ; its sign is that of (8) by (5).

Conversely, reconstruct x^* and k from v, h . The reduced matrix is Hurwitz by (7). Equations (5) and (8) give a negative signed principal minor of J , so Theorem 3.1 supplies $D \succ 0$ for which $J - D$ has a positive real eigenvalue. On $(0, \pi)$ this is the first nonzero Neumann mode; other domain scales are absorbed into D . Conservation laws permit every nonzero-mode amplitude, as shown in Section 2. The full set cannot be negative: its determinant is zero in the conservative case and has positive signed determinant in full rank. $\square$

Let the characteristic polynomial of $R(v, h)$ be $\lambda^s + a_1 \lambda^{s-1} + \dots + a_s$. The Routh–Hurwitz determinants $\Delta_1, \dots, \Delta_s$ turn (7) into strict polynomial inequalities.

Corollary 3.3 (Existential-real membership). *The arbitrary-network decision problem belongs to the existential theory of the reals and has a polynomial-size formula computed from (Γ, Y) .*

Proof. Introduce Boolean variables $b_i^2 - b_i = 0$ and $S_b = \text{diag}(b_1, \dots, b_n)$. If $I = \{i : b_i = 1\}$, simultaneous permutation of rows and columns gives

$$\det(I - S_b - S_b A(v) S_b) = (-1)^{|I|} \det A(v)[I, I]. \quad (9)$$

The empty selector gives 1 and cannot satisfy negativity. Determinants, characteristic coefficients, and Hurwitz determinants have polynomial-size arithmetic circuits. Rational row reduction produces B, C with polynomial encoding length, and gate variables convert the circuits to a polynomial-size existential-real formula. This proves membership only, not $\exists\mathbb{R}$ -hardness. $\square$

4 NP-hardness

We reduce from PARTITION. The source matrix family follows Blondel and Tsitsiklis [3]; all additional lifting, scaling, and mass-action realization steps are stated below.

Given positive integers $a_1, \dots, a_\ell$, choose $k = \lceil \sqrt{\ell + 1} \rceil$, put $m = k^2$, and pad to $a \in \mathbb{Z}^m$. Let

$$\gamma = a^T a, \quad Q = I_m + aa^T, \quad \beta = 1 - \frac{1}{2m(1 + \gamma)},$$

and

$$B(x, y) = \begin{pmatrix} -kQ & y \\ x^T & k\beta \end{pmatrix}, \quad x, y \in (-1, 1)^m. \quad (10)$$

Lemma 4.1 (Open-cube partition family). *The family (10) contains a Hurwitz matrix if and only if the PARTITION instance is YES.*

Proof. If $t \in \{\pm 1\}^m$ and $a^T t = 0$, set $r = (1 + \beta)/2$, $x = rt$, and $y = -rt$. The subspace spanned by a has eigenvalue $-k(1 + \gamma)$, the complement orthogonal to a, t has eigenvalue $-k$, and the remaining block is

$$\begin{pmatrix} -k & -rk \\ rk & k\beta \end{pmatrix}.$$

Its trace is negative and determinant $k^2(r^2 - \beta) = k^2(1 - \beta)^2/4 > 0$.

Conversely, interpolate from $(x, y) = (0, 0)$, where there is exactly one positive eigenvalue, to a Hurwitz endpoint. The signed determinant changes sign, so $B(\theta x, \theta y)$ is singular for some $0 < \theta < 1$. With $z = -y$, the Schur complement gives

$$\theta^2 x^T Q^{-1} z = m\beta,$$

hence $x^T Q^{-1} z > m\beta$. Positive definiteness gives

$$\max_{x, z \in [-1, 1]^m} x^T Q^{-1} z = \max_{t \in \{\pm 1\}^m} t^T Q^{-1} t.$$

Using $Q^{-1} = I - aa^T/(1 + \gamma)$, this maximum equals

$$m - \frac{\min_t (a^T t)^2}{1 + \gamma}.$$

Since $m\beta = m - 1/[2(1 + \gamma)]$, the integer $a^T t$ must vanish. $\square$

Write the varying entries as a rank-one affine family $B(q) = B_0 + UX(q)$, where row r of $X(q)$ is $q_r w_r^T$, and define

$$L(q) = \begin{pmatrix} B_0 & U \\ X(q)(B_0 + I) & X(q)U - I \end{pmatrix}. \quad (11)$$

Lemma 4.2 (Row splitting and right-scaling elimination). *The lower-unitriangular matrix $P(q) = \begin{pmatrix} I & 0 \\ -X(q) & I \end{pmatrix}$ satisfies*

$$P(q)L(q)P(q)^{-1} = \begin{pmatrix} B(q) & U \\ 0 & -I \end{pmatrix}.$$

Moreover,

$$\exists q \in (-1, 1)^{2m}, \exists D \succ 0 : L(q)D \text{ Hurwitz}$$

if and only if the partition instance is YES.

Proof. The similarity follows by direct block multiplication. A YES witness uses $D = I$. Conversely, write $D = \text{diag}(D_0, D_1)$. At $q = 0$ the matrix is block upper triangular. Its $-kQD_{0,t}$ block is similar to a negative definite matrix, the scalar $k\beta d_*$ is positive, and $-D_1$ is negative definite. Thus $L(0)D$ has exactly one positive eigenvalue. A Hurwitz endpoint has the opposite signed determinant, so some $L(\theta q)D$ with $0 < \theta < 1$ is singular. Since D is nonsingular and $\det L(q) = (-1)^{2m} \det B(q)$, the original matrix $B(\theta q)$ is singular. The strict cube argument in Theorem 4.1 then forces a partition. $\square$

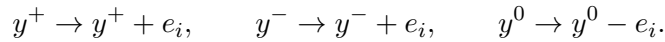
The lift has $m + 1$ fixed rows and $2m$ rows of the form $a_i + q_i b_i$.

Lemma 4.3 (Exact mass-action row realization). *For any rational independent-row family $L(q)$ with fixed and affine one-parameter rows, a classical mass-action network on the same number of species has complete positive steady-flux image*

$$\{\Gamma \text{diag}(v)Y^T : v > 0, \Gamma v = 0\} = \{RL(q) : R \succ 0, q \in (-1, 1)^d\}.$$

It has full stoichiometric rank and uses two reactions per fixed row and three per variable row.

Proof. For a variable row choose an integer c clearing denominators of $a \pm b$ and a sufficiently large integer base complex y^0 with the row species present. Let $y^\pm = y^0 + c(a \pm b)$ and use reactions



Positive steady fluxes have the exact form $u, w, u + w$. Their row contribution is

$$c((u + w)a + (u - w)b) = \rho(a + qb), \quad \rho = c(u + w) > 0, \quad q = \frac{u - w}{u + w} \in (-1, 1).$$

Every ρ, q is recovered by $u = \rho(1 + q)/(2c)$, $w = \rho(1 - q)/(2c)$. A fixed row uses the analogous $+e_i, -e_i$ pair. Because each row uses only its own stoichiometric direction, $\Gamma v = 0$ contains exactly these rowwise equations and no cross-row constraints. Each direction e_i occurs, so the rank is full. $\square$

Theorem 4.4 (NP-hardness). *Weak all-mobile stationary-Turing admissibility is NP-hard under polynomial-time many-one reductions from PARTITION. An instance of length ℓ produces a full-rank network with $3m + 1$ species and $8m + 2$ reactions, where $m = k^2 > \ell$, and every stoichiometric column is $\pm e_i$. All rates and equilibrium coordinates in a YES witness are strictly positive. The reaction molecularity is not uniformly bounded.*

Proof. Apply Theorem 4.3 to $L(q)$. Every positive-equilibrium Jacobian is

$$J = RL(q)H,$$

which is similar to $L(q)HR$. By Theorem 4.2, such a Jacobian is Hurwitz exactly for YES partition instances. The coordinate inherited from the bottom-right entry of B satisfies $L_{jj} = k\beta > 0$, so every realized Jacobian has $J_{jj} > 0$ and hence a negative signed singleton minor. A Hurwitz realization is therefore stationary-destabilizable by Theorem 3.1; the reverse implication includes homogeneous stability by definition.

There are $2(m+1) + 3(2m) = 8m + 2$ reactions. All rational entries, denominator clearing factors, and nonnegative base complexes have polynomial bit length. A YES witness takes the rational open-cube point, $R = H = I$, $x^* = \mathbf{1}$, and rates equal to the positive row fluxes. $\square$

This theorem does not imply that no compact criterion can exist. It implies that polynomial-time recognition is impossible unless $P = NP$; a compact condition could itself be computationally hard to evaluate.

5 Polynomial-time algorithms for fixed species count

Define

$$K = \{v \in \mathbb{R}_{\geq 0}^m : \Gamma v = 0\}, \quad \Phi(v) = \Gamma \operatorname{diag}(v) Y^T, \quad P = \Phi(K) \subseteq \mathbb{R}^{n \times n}.$$

Lemma 5.1 (Positive fluxes project to the relative interior). *If K contains a strictly positive vector, then*

$$\{\Phi(v) : v > 0, \Gamma v = 0\} = \operatorname{ri} P.$$

Proof. The strict point ensures that no coordinate inequality is identically tight, so $\operatorname{ri} K = K \cap \mathbb{R}_{> 0}^m$. Linear maps preserve relative interiors of nonempty convex sets [12]. $\square$

Theorem 5.2 (Fixed-species polynomial time). *For every fixed n , weak all-mobile stationary-Turing admissibility of rationally encoded classical mass-action networks is decidable in polynomial bit complexity in the reaction count and input length. The same holds when a designated mobile set is included in the input.*

Proof. Strict positive-flux feasibility is a rational linear program: by homogeneity, it is equivalent to $\Gamma v = 0$ and $v_r \geq 1$. Every extreme ray of K is a support-minimal nonnegative dependency among columns of Γ , hence has support at most $\operatorname{rank} \Gamma + 1 \leq n + 1$. For fixed n , enumerate all such column subsets, compute their exact one-dimensional positive kernels, and map the resulting circuit rays through Φ . There are $O(m^{n+1})$ tests and the generators have polynomial bit length by determinant bounds.

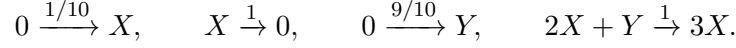
These images generate P in fixed ambient dimension n^2 . Exact fixed-dimensional polyhedral algorithms compute its span, lineality, and a relative-facet description in polynomial time [15]. Thus $A \in \operatorname{ri} P$ is described by linear equations and strict rational inequalities, including lower-dimensional or nonpointed images.

By Theorem 5.1, the remaining formula uses only the n^2 entries of A and n positive entries of h : relative-facet conditions, Routh–Hurwitz inequalities for $A \operatorname{diag}(h)$ on S , and one negative signed principal minor. For fixed n , the variable count and degrees are fixed and the number and bit length of coefficients are polynomial. Fixed-variable algorithms for the first-order theory of the reals therefore run in polynomial bit complexity [11, 2]. A designated mobile set adds one spectral variable and one of constantly many block choices. $\square$

The conclusion is a polynomial-time exact decision algorithm, not a finite structural classification. In particular it applies to arbitrary reaction lists for $n = 1, 2, 3$.

6 A biochemical two-species benchmark

Consider the Schnakenberg reactions



The mass-action equations are

$$\dot{x} = \frac{1}{10} - x + x^2y, \quad \dot{y} = \frac{9}{10} - x^2y.$$

At $(x^*, y^*) = (1, 9/10)$ the positive flux is $(1/10, 1, 9/10, 9/10)$ and the Jacobian is

$$J = \begin{pmatrix} 4/5 & 1 \\ -9/5 & -1 \end{pmatrix}.$$

Its trace is $-1/5$ and determinant is 1, so it is Hurwitz. The positive entry J_{11} is a negative signed singleton minor. Indeed, with

$$D = \text{diag}(1/100, 10), \quad \det(D - J) = -\frac{689}{100} < 0,$$

$J - D$ has a positive real eigenvalue. This example illustrates the exact network formula in the classical activator–inhibitor setting; it does not assert a nonlinear patterned branch.

7 Data level, designated mobility, and certificates

The exact formula uses the indexed pair (Γ, Y) . Among the six levels considered—unsigned interaction graph, signed interaction graph, a Coates graph at one equilibrium, Γ alone, (Γ, Y) , and the fully labeled indexed complex graph—Level 5 is the weakest sufficient parameter-independent input. Exact separating network pairs show that Levels 1 and 2 can have the same signs but opposite answers and that the same ordered Γ with different sources can have opposite answers. Level 3 depends on a selected parameter realization. Levels 5 and 6 are equivalent when reaction indices and integer labels are retained, since $y'_r = y_r + \gamma_r$. This statement is only relative to these six representations.

If only species in M diffuse, the determinant-at-zero principal-minor rule is false. The correct fixed- J statement uses a positive spectral parameter.

Theorem 7.1 (Designated mobile species). *Assume $\det(\lambda I - J) > 0$ for every $\lambda > 0$. There is a diagonal matrix positive on M and zero on M^c such that $J - D$ has a positive real eigenvalue if and only if there are $\lambda > 0$ and $I \supseteq M^c$ with*

$$\det(\lambda I_{|I|} - J[I, I]) < 0.$$

Proof. Expanding only in mobile diagonal variables gives

$$\det(\lambda I - J + D_M) = \sum_{I \supseteq M^c} \det(\lambda I_{|I|} - J[I, I]) \prod_{j \in M \setminus I} d_j.$$

A zero with positive background coefficient requires a negative coefficient. Conversely, the same multiscale assignment used in Theorem 3.1 uniquely dominates a selected negative coefficient; the determinant is then negative at the chosen λ and positive for large spectral parameter, yielding a larger positive root. For mass-action homogeneous stability, (2) supplies the hypothesis. $\square$

An exact counterexample to the zero-parameter substitute is

$$J = \begin{pmatrix} -10 & 20 & -50 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}, \quad M = \{1\}.$$

The matrix is Hurwitz and the relevant zero-parameter signed block test is nonnegative, but $J - \text{diag}(257/2, 0, 0)$ has eigenvalue $3/2$; at $\lambda = 3/2$ the immobile block determinant is $-1/4$.

Theorem 7.2 (Finite exact certificates). *Every YES instance has a finite exact real-algebraic sample certificate. Every NO instance has a finite exact Real-Nullstellensatz certificate. Both are independently checkable. No polynomial certificate-size bound or practical general certificate generator is claimed.*

Proof sketch. Replace $g > 0$ by $gz^2 - 1 = 0$ and $g < 0$ by $(-g)z^2 - 1 = 0$. A nonempty rational semialgebraic set has a point with real-algebraic coordinates, representable in one isolated number field. If a branch has no real zero, the Real Nullstellensatz gives

$$-1 = \sum_j s_j^2 + \sum_k q_k F_k.$$

Exact coefficient comparison verifies the identity. There are finitely many principal branches. Detailed formats and generation/size distinctions are in the supplement [4, 2]. $\square$

8 Discussion

The exact characterization exposes the two coupled parts of the network-wide problem: positive steady-flux geometry and positive right-diagonal homogeneous stabilization. The NP-hardness theorem shows that their interaction already encodes a classical combinatorial decision problem even when every stoichiometric column is a unit coordinate change. For fixed species count, the flux variables can nevertheless be projected away into fixed matrix dimension, producing polynomial-time exact algorithms.

Several directions remain open. Most important biologically is whether the hardness survives bimolecular or uniformly bounded-molecularity networks. The present denominator-clearing construction does not. Strong NP-hardness and $\exists\mathbb{R}$ -hardness are also unknown. At the dynamical level, weak stationary admissibility has not been shown equivalent to a primary simple transverse stationary crossing, and no arbitrary-dimensional wave criterion or nonlinear patterned-state theorem is supplied.

The accompanying software contains an independent constructor, verifier, and adversarial tests. It does not bundle a complete general-purpose quantifier-elimination engine: a raw network without an attached exact certificate may remain unresolved computationally even though the abstract decision procedure terminates.

Computational assistance and reproducibility. Symbolic computation, numerical falsification, and AI-assisted drafting were used in developing and auditing the work. The mathematical claims in this version are supported by explicit proofs and exact independently replayable artifacts; numerical searches are not used as proofs.

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